

ALEX LAGUË

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Research Experience

University of Pennsylvania & Center for Particle Cosmology Postdoctoral Fellow	Philadelphia, USA <i>2022 to Present</i>
University of Toronto & Canadian Institute for Theoretical Astrophysics Graduate Research Assistant	Toronto, Canada <i>2017-2022</i>
McGill University Undergraduate Researcher (Mathematics) Undergraduate Researcher (Physics)	Montréal, Canada <i>2016</i> <i>2015 & 2017</i>

Education

University of Toronto & Canadian Institute for Theoretical Astrophysics PhD in Astronomy & Astrophysics Thesis: Ultralight Axions and the Cosmic Web Supervisors: J. Richard Bond, Renée Hložek	Toronto, Canada <i>2017 to 2022</i>
McGill University BSc in Honours Mathematics and Physics First Class Honours Thesis: Cross-Correlations to Probe the Existence of Cosmic Strings Supervisor: Robert Brandenberger	Montréal, Canada <i>2014-2017</i>

Awards and Fellowships

Ontario Graduate Scholarship.	2021-2022
Frank S. Hogg Memorial Fellowship.	2020
Mary and Ron Martin Graduate Fellowship in Astrophysics.	2018
Natural Sciences and Engineering Research Council of Canada Undergraduate Research Award.	2016
Fonds de recherche Québec, Nature et Technologie Undergraduate Research Award.	2016

Collaboration Memberships

Atacama Cosmology Telescope and Simons Observatory

Peer-Review Referee

Monthly Notices of the Royal Astronomical Society, Physical Review D, Astronomy and Astrophysics, Annalen der Physik.

Selected Presentations

- First Measurement of 3-Dimensional Reconstructed Velocity Fields from the Kinetic Sunyaev-Zeldovich Effect. New Physics from Old Light, Cambridge University (2024).
- Probing the String Axiverse with Large-Scale Structure, Seminar, UCLA (2024).
- Dark Matter Under the Lens. Future Science with CMB x LSS. YITP, Kyoto (2023).
- Constraining Ultralight Axions with Large-Scale Structure. New Horizons in Astro and Particle Theory, Queens University (2021).
- Efficient Modelling of Ultralight Axions. LEPP Seminar, Cornell (2021).
- Introduction to Fuzzy Dark Matter. From Inflation to the Hot Big Bang, Kavli Institute for Theoretical Physics, Santa-Barbara, USA (2020).
- Non-Linear Fuzzy Dark Matter Modelling with Extended LPT. The CMB in HD: The Low-noise High-resolution Frontier, New York, USA (2019).

Teaching Experience

Guest lecturer for the following classes: Topics in Cosmology

Teaching assistant (TA) for the following classes: AST 101: The Sun and Its Neighbors[†], AST 201: Stars and Galaxies[†], AST 210: Great Moments in Astronomy[†], AST 222: Galaxies and Cosmology*, AST 325: Introduction to Practical Astronomy*, AST 326: Practical Astronomy*.

* Special appointment as computing teaching assistant involving the creation of a Python guide and tutorials for astronomy students, taking them from their first line of code to data analysis for research purposes.

[†] Included preparing and leading tutorials and exam review sessions as well as hosting office hours (both in-person and virtual).

Outreach, Mentoring and Scientific Communication

- Astro on Tap Philadelphia
- Panel moderator for the general public symposium *Mysteries of the Universe: Black Holes, Dark Matter, and Dark Energy* organized by the Astronomy and Space Exploration Society.
- Planetarium operator which involved creating and presenting planetarium shows to the general public about recent events in astronomy.
- Animator for the public outreach event *AstroTours* which involved presenting detailed 3D-printed models of existing and upcoming telescopes to the general public.
- Student Mentorship: Karen S. Perez (graduate student), Zara Zaman (undergraduate), Alexander Spencer London (undergraduate) and Ishika Bangari (mentorship program).

Skills

Python, JAX, Numba, Parallel & GPU Programming, High Performance Computing.